

1855 ▶ 1865

1856

Artificial dye

British chemist William Perkin was still a student when he accidentally discovered the first synthetic chemical dye. It was an intense purple color, which became known as mauve (the French name of the purple mallow flower) and was later given the chemical name mauveine. Perkin set up a factory to produce mauve commercially.

Dress made of silk dyed with Perkin's mauve

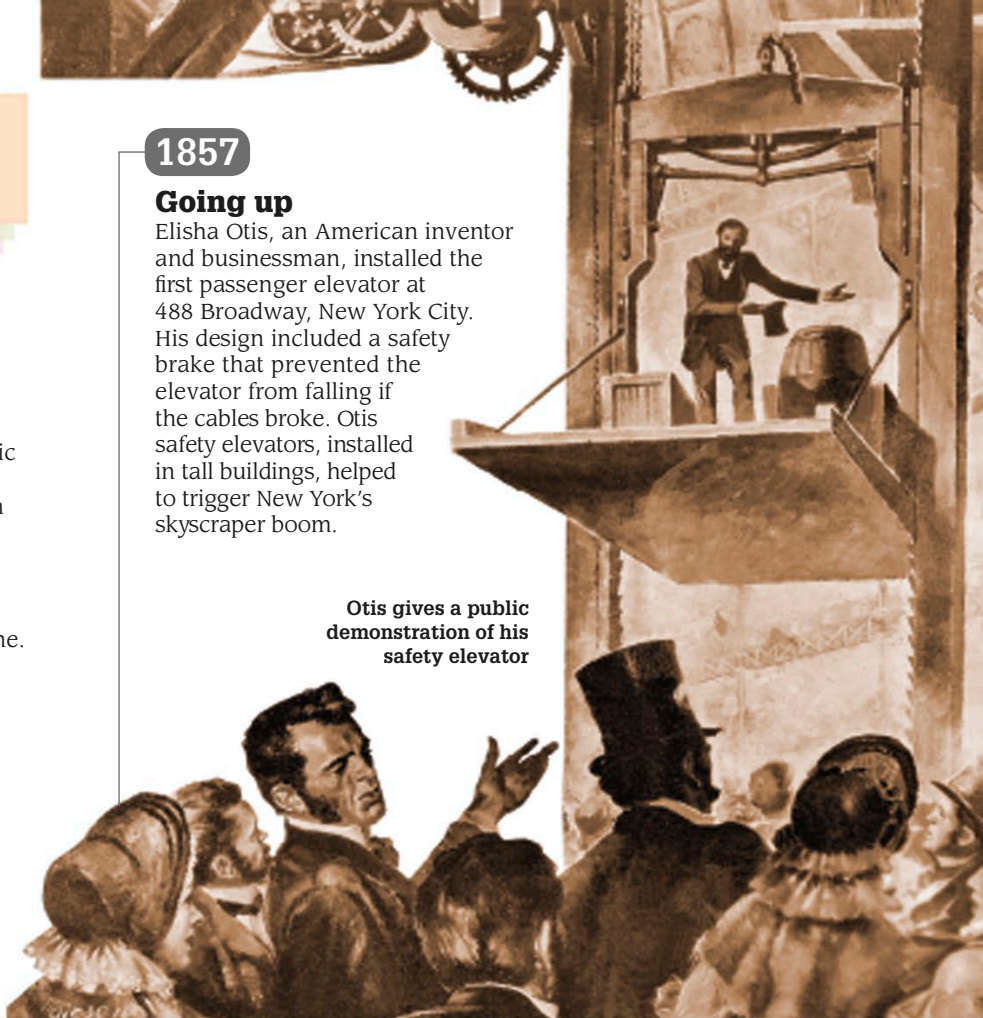


1857

Going up

Elisha Otis, an American inventor and businessman, installed the first passenger elevator at 488 Broadway, New York City. His design included a safety brake that prevented the elevator from falling if the cables broke. Otis safety elevators, installed in tall buildings, helped to trigger New York's skyscraper boom.

Otis gives a public demonstration of his safety elevator



1855

1856

Steelmaking process

British engineer Henry Bessemer patented a furnace for making steel cheaply by blowing air through molten iron to remove the impurities. Steel rails for railroads, made by the Bessemer process, lasted ten times longer than iron ones.

1856

Studying pea plants

Austrian monk Gregor Mendel began breeding pea plants. He discovered that inherited characteristics, such as size, are passed on by invisible "factors." We now call them genes. Mendel's work, unrecognized at the time, forms the basis of modern genetics.



1859

Influential book

British naturalist Charles Darwin published a book called *On the Origin of Species*, in which he explained his theory of evolution through natural selection. It is one of the most important books in the history of science.



1858

Transatlantic cable

The first undersea transatlantic telegraph cable, between western Ireland and Newfoundland (off the coast of Canada), came into service, but it failed after three weeks and had to be abandoned. A new, longer-lasting cable was successfully laid between 1865-1866.

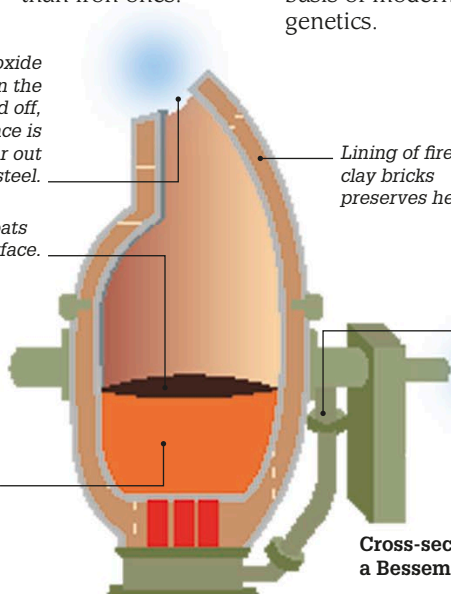
4. Carbon monoxide gas produced in the process is burned off, and the furnace is tilted to pour out the steel.

3. Slag floats to the surface.

2. Oxygen in the air combines with carbon (to produce carbon monoxide), and with manganese and other impurities, creating slag.

Lining of fire clay bricks preserves heat.

1. Air entering through holes at the bottom of the furnace is blown through molten iron.



Cross-section through a Bessemer furnace